Immunological aspects of hormones

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3rd Stage

IBSII / LG 7

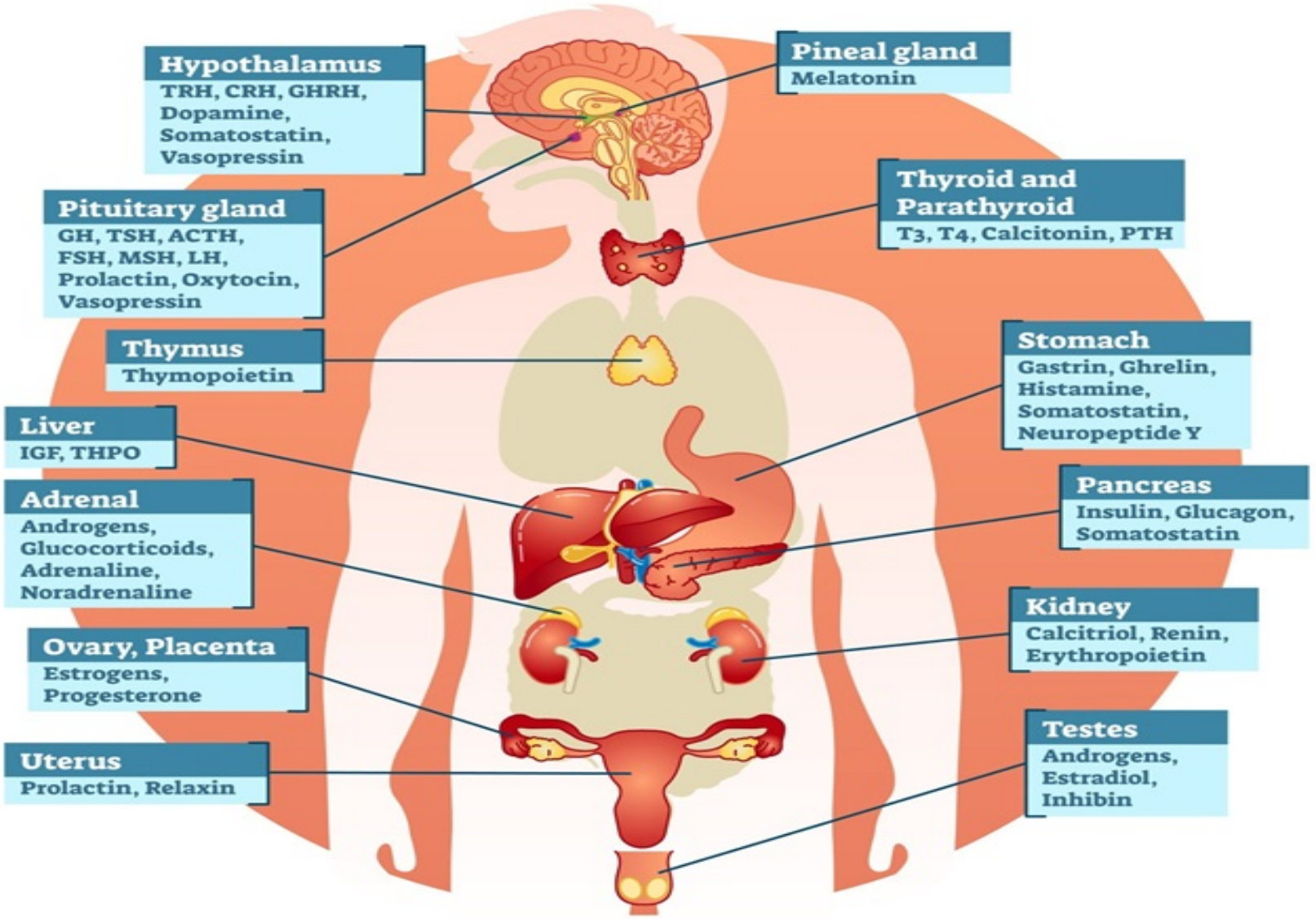
Background

- A hormone is a chemical substance that is secreted into the body fluids (blood) by one cell or a group of cells (gland) and has a physiological control effect on other cells (targets) of the body.

- In humans, hormones are produced mainly by the endocrine glands and many other organs and tissues that contain endocrinocytes (secrete specific hormones).

❑ Hormones act by binding to specific chemical receptors on cell membranes or within cells.

HORMONES


Hypothalamus
TRH, CRH, GHRH, Dopamine, Somatostatin, Vasopressin
Pineal gland
Melatonin
Pituitary gland
GH, TSH, ACTH, FSH, MSH, LH, Prolactin, Oxytocin, Vasopressin
Thyroid and Parathyroid
T3, T4, Calcitonin, PTH
Thymus
Thymopoietin
Stomach
Gastrin, Ghrelin, Histamine, Somatostatin, Neuropeptide Y
Liver
IGF, THPO
Adrenal
Androgens, Glucocorticoids, Adrenaline, Noradrenaline
Pancreas
Insulin, Glucagon, Somatostatin
Kidney
Calcitriol, Renin, Erythropoietin
Ovary, Placenta
Estrogens, Progesterone
Testes
Androgens, Estradiol, Inhibin
Uterus
Prolactin, Relaxin

Abbreviation	Full Name
TSH	Thyroid-Stimulating Hormone
MSH	Melanocyte-Stimulating Hormone
GH	Growth Hormone
IGF	Insulin-like Growth Factor
THPO	Thrombopoietin
T3	Triiodothyronine
T4	Thyroxine (Tetraiodothyronine)
PTH	Parathyroid Hormone
TRH	Thyrotropin-Releasing Hormone

- The major endocrine glands which are scattered throughout the body

- Hormones help in body growth, development, metabolism, sexual function, reproduction etc.

- Some glands also have non-endocrine regions that have functions other than hormones secretion.

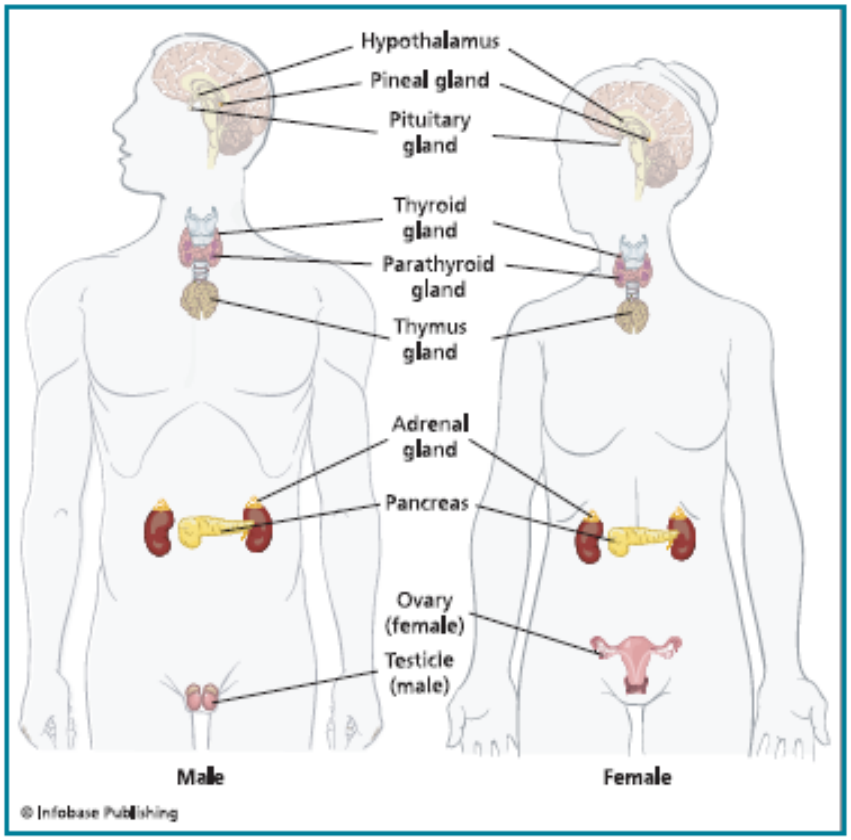


Figure 3.1 Each of the endocrine glands produces one or more hormones. Some organs, like the pancreas and kidneys, also have other functions that are not related to hormones.

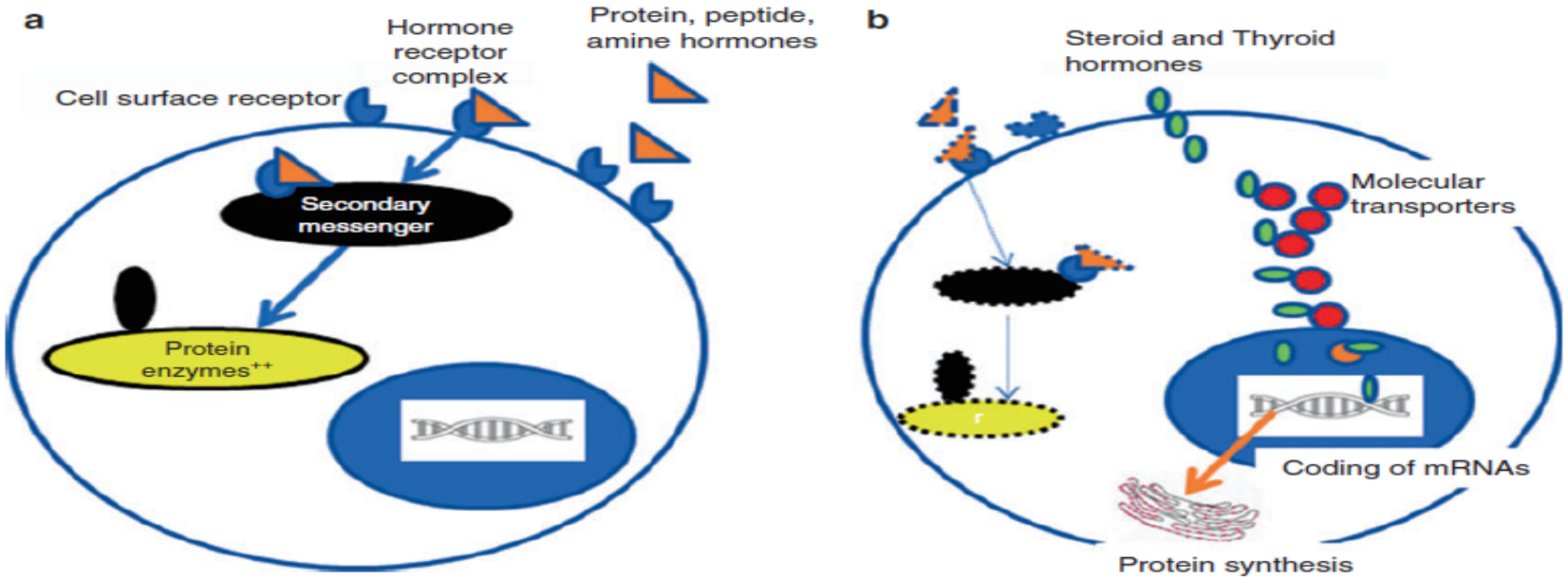
Pancreas → exocrine portion that secretes digestive enzymes.

Pancreas → endocrine portion that secretes hormones.

Hormones

- Non steroid
- Produced proteins, small peptides and amino acid derivative
- Glucagon, insulin, prolactin, thyroid hormones

- Steroid
- Produced from cholesterol
- Sex hormones(estrogen, progesterone, testosterone)



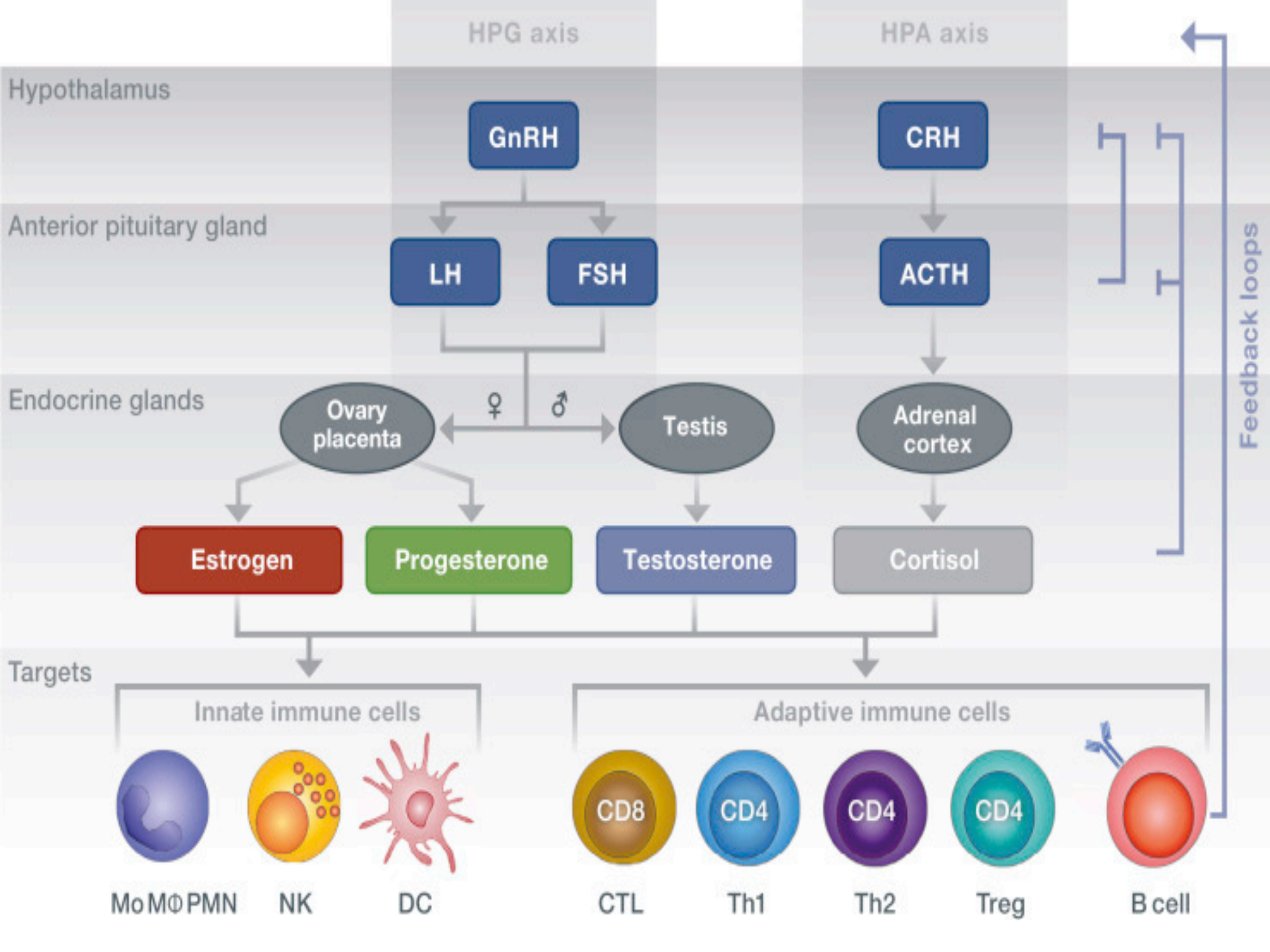
Endocrine System, Fig. 7 Mechanisms of action (a) Protein, peptide, and amine hormones (b) Steroid and thyroid hormones. (Reproduced with permission from Hormones and Behavior, Kumar A. et al., 2018, Encyclopedia of Animal Cognition and Behavior, Springer Nature)

Hormone and immune system

- The immune system is highly receptive to endocrine signals due to the expression of hormone receptors on immune cells.

- Endocrine tissues have been shown to be responsive to immune mediators, such as cytokines.

- The significant impact of hormones on the immune system becomes evident in the differential susceptibility to and progression of diseases that can be observed in men and women.


HPG axis
HPA axis
Hypothalamus
GnRH
CRH
Anterior pituitary gland
LH
FSH
ACTH
Endocrine glands
Ovary placenta
♀
♂
Testis
Adrenal cortex
Estrogen
Progesterone
Testosterone
Cortisol
Feedback loops
Targets
Innate immune cells
Adaptive immune cells
CD8
CD4
CD4
CD4
Mo MΦ PMN
NK
DC
CTL
Th1
Th2
Treg
B cell

The Major Neuroendocrine Axes

- To understand how hormones influence immunity, we need to know the pathways through which they act. The two most important are the **HPG axis** and the **HPA axis**.
- **The HPG Axis (Hypothalamic-Pituitary-Gonadal Axis)**
- This axis controls the production of sex hormones:
- The **hypothalamus** releases **GnRH** (gonadotropin-releasing hormone).
- GnRH stimulates the **anterior pituitary** to release **LH** (luteinizing hormone) and **FSH** (follicle-stimulating hormone).
- LH and FSH act on the **gonads** — the ovaries in females and the testes in males.
- The gonads produce **estrogen**, **progesterone**, and **testosterone**.

The Major Neuroendocrine Axes

- **The HPA Axis (Hypothalamic-Pituitary-Adrenal Axis)**
- This axis controls the production of **cortisol**, the body's primary stress hormone:
- The **hypothalamus** releases **CRH** (corticotropin-releasing hormone).
- CRH stimulates the **anterior pituitary** to release **ACTH** (adrenocorticotropic hormone).
- ACTH acts on the **adrenal cortex** to produce **cortisol**.
- Cortisol is a powerful **immunosuppressive** hormone. It is used clinically to treat inflammatory and autoimmune conditions. But it also plays a role in normal immune regulation.

Estrogen

- Estrogens exert their effect via intracellular estrogen receptors ERα and ERβ.
- Estrogen receptors are expressed in various types of immune cells, including lymphocytes, macrophages, and DC.
- Estrogens have both pro- and anti-inflammatory roles, depending on several factors including:-

a. Type of the immune cell target
b. Concentration of the hormone
c. Type of immune stimulus (foreign or auto-antigens).
d. Relative expression of ERα and ERβ.

- Estrogens can enhance cell-mediated and humoral immune responses.
- Estrogens have antioxidant properties
- Estrogens may protect immune cells against apoptosis .
- Estrogen also promotes mammary, ovarian and endometrial tumorigenesis.

Progesterone

- Its receptors s have been identified in epithelial cells, mast cells, granulocytes, macrophages, and lymphocytes.
- It is typically known as an immunosuppressive agent:-

- It inhibits the activation of NFκB.
- Reduces macrophage, DCs and NK cell activity.
- It decreases Ab-secreting B cell numbers and suppresses their maturation.
- Inhibits proliferation of T cells after DC stimulation and their differentiation into proinflammatory Th17 cells.

❑ Elevated concentrations of progesterone during pregnancy:-

- Inhibit development of Th1 responses.
- Inhibit production of proinflammatory cytokines such as IFN-*γ.*
- Stimulate the generation of Treg cells.
- Promoting Th2 immune responses, including the synthesis of anti-inflammatory cytokines such as IL-4, and IL-10.

Testosterone

✓ Testosterone exerts immune-suppressive effects on the immune response, which has been suggested to protect males from autoimmunity.

✓ Suppresses DCs activation via.

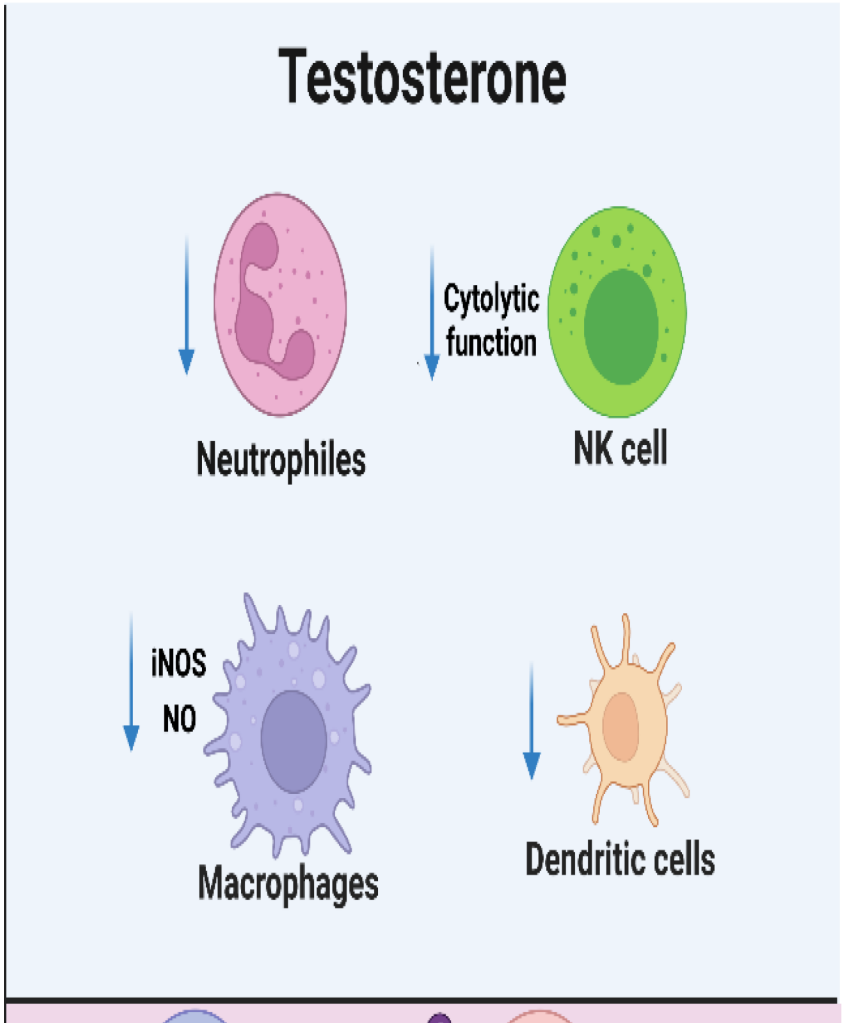
- Inhibition of costimulatory molecule expression.
- Reduced proinflammatory cytokine production.
- Diminished antigen-responsive capacity.

✓ Suppresses NK cell proliferation.

Innate immune response

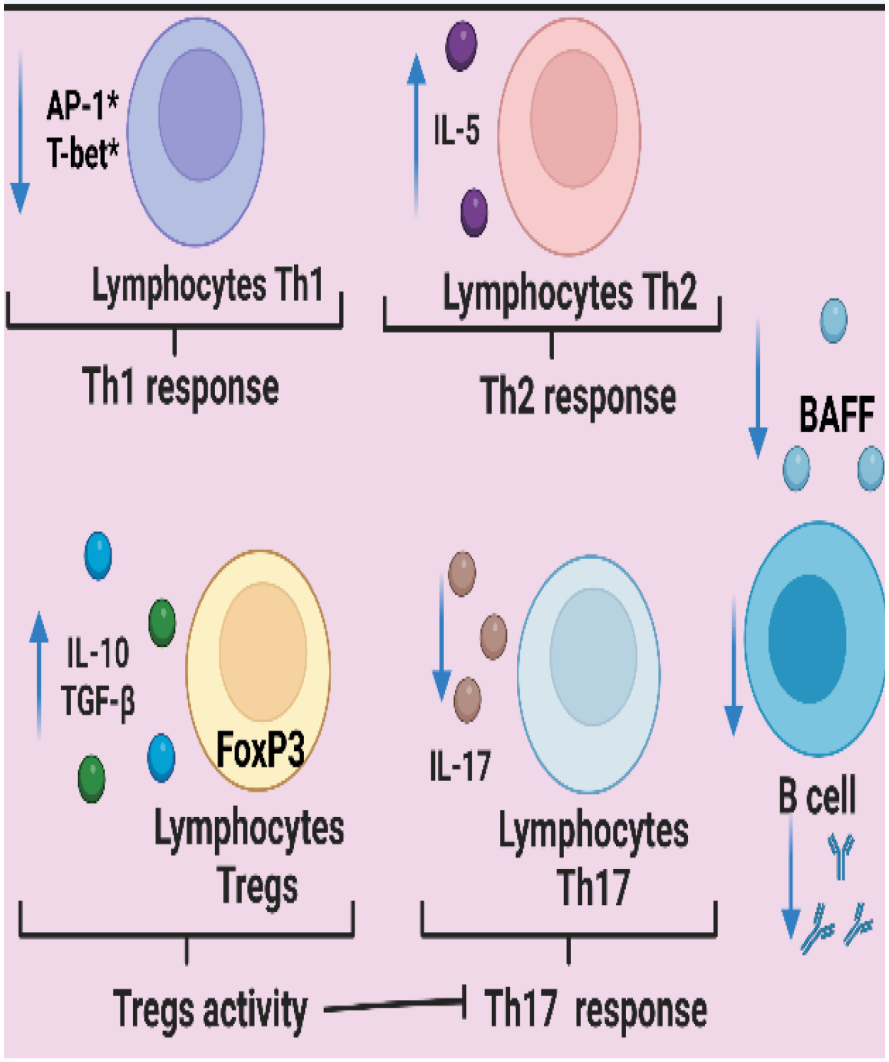
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- Testosterone decreases neutrophil proliferation,
- NK-cell cytolytic activity,
- and macrophage nitric oxide (NO) synthesis by decreasing iNOS expression and
- dendriti cell proliferation



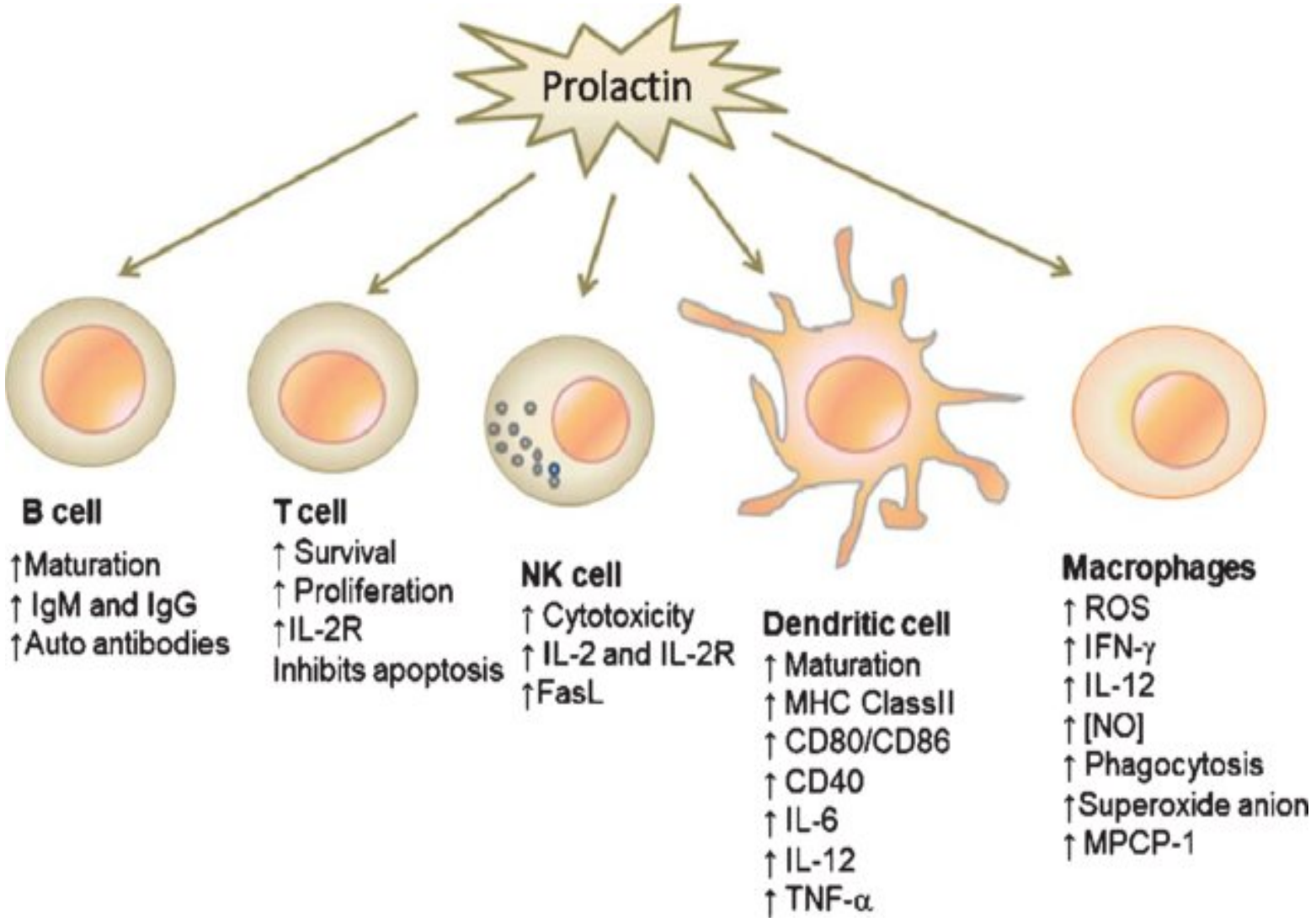
Adaptive immune response

- ✓ Reduces T cell proliferation and cytokine production.
- ✓ Testosterone inhibits Th1 differentiation, IFN-γ secretion in NKT cells and parasite-induced Th2 responses.
- ✓ Enhances Th2 immunity to counteract overshooting Th1 responses in autoimmunity, and induces expansion of Tregs.
- ✓ Testosterone strengthens tight junctions of the blood–testis barrier, thereby supporting the immune privilege of the testis



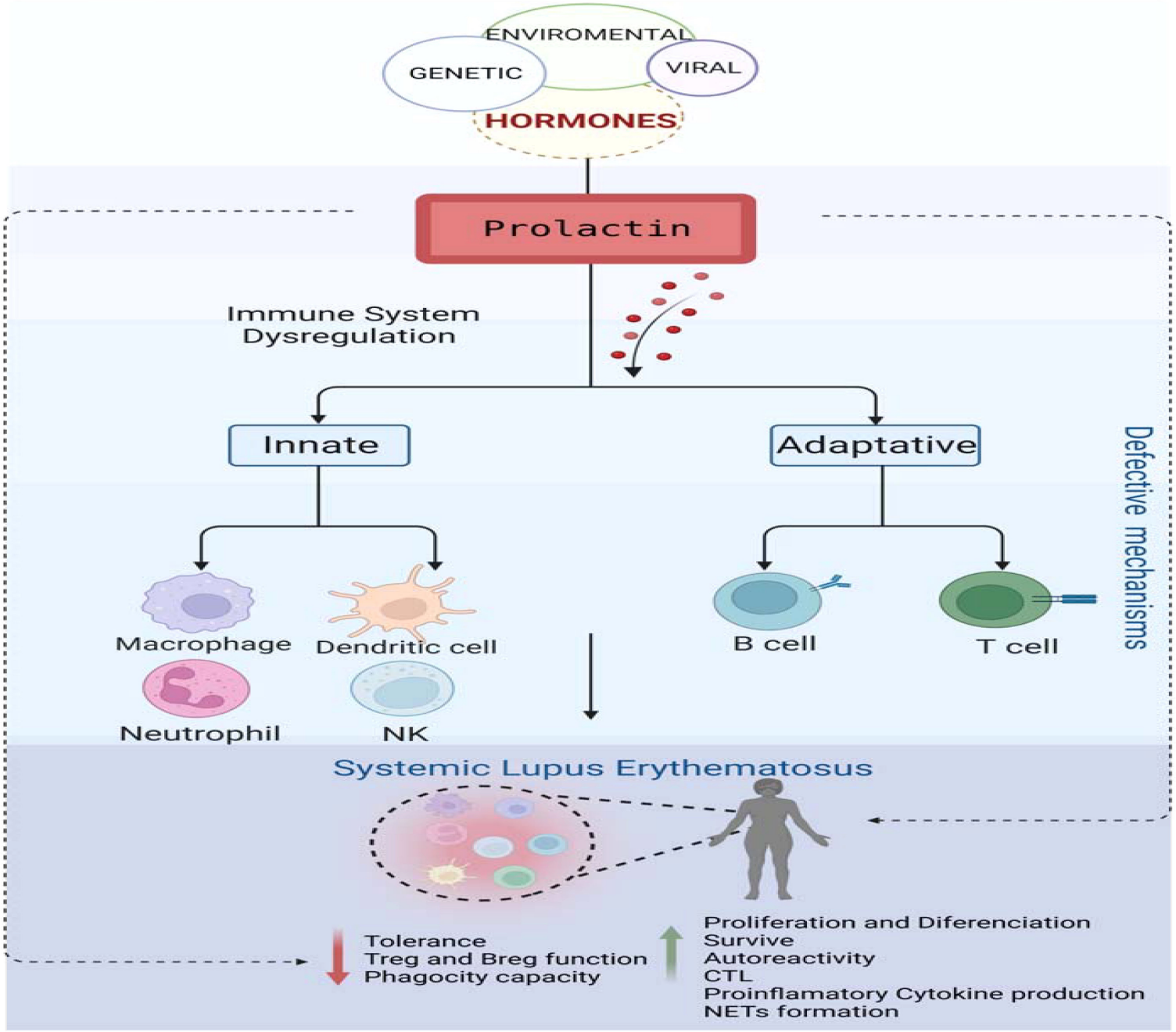
Prolactin (PRL)

- Is acting as a protein hormone, as well as a cytokine with pleiotropic function.
- It is produced by many immune cells like macrophages, NK cells, and T- and B-lymphocytes. As well its receptors widely expressed.
- PRL secretion is regulated by cytokines such as interleukin (IL)-1, IL-2, and IL-6, which are stimulators, while endothelin-3 and interferon (IFN)-γ play an inhibitory role.
- PRL has pro-inflammatory and anti-inflammatory functions.
- hyperprolactinemia can impair B-cell clonal deletion…..promoting autoimmunity?


Prolactin
B cell
↑Maturation
↑ IgM and IgG
↑Auto antibodies
T cell
↑ Survival
↑ Proliferation
↑IL-2R
Inhibits apoptosis
NK cell
↑ Cytotoxicity
↑ IL-2 and IL-2R
↑FasL
Dendritic cell
↑ Maturation
↑ MHC ClassII
↑ CD80/CD86
↑ CD40
↑ IL-6
↑ IL-12
↑ TNF-α
Macrophages
↑ ROS
↑ IFN-γ
↑ IL-12
↑ [NO]
↑ Phagocytosis
↑Superoxide anion
↑ MPCP-1

- In the physiology of the immune system, PRL acts by stimulating the secretion of other cytokines and the expression of cytokine receptors, and also as a growth and survival factor.

- PRL It is capable of changing Th1 and Th2 type cytokine production, promoting IL-6 and INF-γ secretion, and playing a regulatory role on IL-2 levels.
- During pregnancy, hyperprolactinemia has been associated with lupus (SLE) anticoagulant, disease activity, and poor outcomes for mother and fetus.
- Higher levels of PRL in serum and synovial fluid of patients with rheumatoid arthritis.


ENVIROMENTAL
GENETIC
VIRAL
HORMONES
Prolactin
Immune System
Dysregulation
Innate
Adaptative
Defective mechanisms
Macrophage
Dendritic cell
B cell
T cell
Neutrophil
NK
Systemic Lupus Erythematosus
Tolerance
Treg and Breg function
Phagocity capacity
Proliferation and Diferenciation
Survive
Autoreactivity
CTL
Proinflamatory Cytokine production
NETs formation

leptin

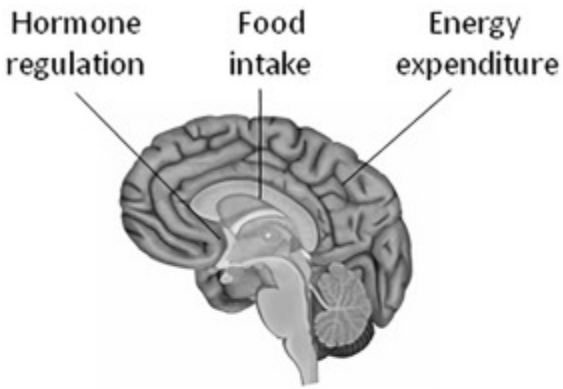
- Leptin together with other regulatory molecules has a central role in appetite and body weight homeostasis.

- Most immune cells express leptin receptor (LEPR) at their surfaces.

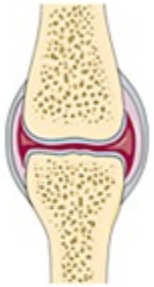
- Malnutrition results in hypoleptinemia, while obesity leads to hyperleptinemia, both conditions affecting the immune response in an opposite manner.

- Obese subjects demonstrated decreased levels of Treg, which are inversely correlated with leptin levels.

- In malnutrition, altered T cell function and metabolism was associated with decreased leptin levels.
- Leptin regulates both innate and adaptive responses.
- In innate immunity, leptin increases the cytotoxicity of NK cells and promotes the activation of granulocytes, macrophages, and DCs.
- In adaptive immunity, leptin increases the proliferation of naïve T and B cells.
- Leptin promotes the switch towards a pro-inflammatory Th1 rather than anti-inflammatory Th2 phenotype., and facilitates Th17 responses.



Central actions



Bone and Cartilage:

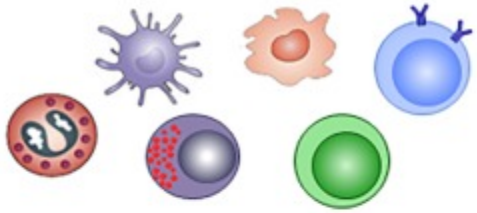
- Bone mass
- Chondrocytes
- Rheumatoid arthritis
- Osteoarthritis



Vascular function:

- Atherosclerosis
- Angiogenesis
- Oxidative stress

Leptin/LEPR



Innate and Adaptive Immune System:

- Cytokine induction
- Chemotaxis
- Natural killer cytotoxicity
- Macrophage activation
- Dendritic cell activation
- Th1 stimulation
- Th2 inhibition
- Naïve T-cell proliferation

Other targets:

- Kidney
- Bowel
- Pancreas
- Muscle

Reproduction:

- Placental function
- Trophoblast invasion
- Foetal function
- Gonadotrophins and gonadal hormones

Pleiotropic nature of leptin

References

- Kumar A, Kulandhasamy M, Kumari C, et al. Endocrine System. In: Vonk J, Shackelford TK, eds. *Encyclopedia of Animal Cognition and Behavior*. Springer Nature; 2018. doi:10.1007/978-3-319-47829-6_483-1
- PR Ashalatha, G Deepa . Textbook of Anatomy & Physiology for Nurses.3rd edition.2011.
- Md. Eaqub Ali Nina Naquiah Ahmad Nizar.Preparation and Processing of Religious and Cultural Foods.2018.
- Ashutosh Kumar, Maheswari Kulandhasamy, Chiman Kumari et al. .Endocrine System. Springer International Publishing AG, part of Springer Nature 2018
- J. Vonk, T. K. Shackelford (eds.), Encyclopedia of Animal Cognition and Behavior,
- . Lynette Rushton, Introduction by Denton A. Cooley. The human body how it works the endocrine system. 2009.
- Ina Annelies Stelzer and Petra Clara Arck .Immunity and the Endocrine System.016
- Vera Francisco,Jesús Pino, Victor Campos-Cabaleiro et al. Obesity, Fat Mass and Immune System: Role for Leptin. Front Physiol. 2018; 9: 640.
- Other from internet sources.